

# The Geometry of Linear Equations

One question, three ways to see it: the **row picture**, the **column picture** and the **matrix form**  $Ax = b$ .

Explained from zero on a real money question: splitting ₹10 lakh between two funds.

2 × 2 and 3 × 3

Portfolio mix

MIT 18.06, Lecture 1

THE QUESTION

You have ₹10 lakh and two funds

| Fund                 | Return per year | On ₹1 lakh   |
|----------------------|-----------------|--------------|
| Nifty 50 index fund  | 12%             | ₹12 thousand |
| Government bond fund | 7%              | ₹7 thousand  |

You want to invest all ₹10 lakh and earn ₹1,05,000 a year.

THE QUESTION

How much goes into each fund?

Returns are illustrative round numbers, not forecasts.

## STEP 1

# Turn it into equations

Let  $x = ₹$  lakh in Nifty and  $y = ₹$  lakh in bonds. Each requirement is one equation:

$$x + y = 10 \quad \text{budget (₹ lakh)}$$

$$12x + 7y = 105 \quad \text{return (₹ thousand)}$$

Two equations that must both hold are a **system**. Each unknown is only multiplied by a number and added, so the equations are **linear**.

### THE FUNDAMENTAL PROBLEM OF LINEAR ALGEBRA

Solve  $n$  linear equations in  $n$  unknowns.

# Three ways to read the same equations

| View             | You look at            | The question becomes                             |
|------------------|------------------------|--------------------------------------------------|
| Row picture      | one equation at a time | Where do the lines <b>meet</b> ?                 |
| Column picture ★ | one fund at a time     | Which <b>mix</b> of fund arrows hits the target? |
| Matrix form      | everything at once     | Find $x$ with $Ax = b$ .                         |

The ★ marks the view the whole course is built on.

## STEP 2

# Each fund is a vector

₹1 lakh of a fund uses ₹1 lakh of budget and earns its return. Stack the two numbers in a column:

$$\text{Nifty} = \begin{bmatrix} 1 \\ 12 \end{bmatrix} \quad \text{Bond} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

- **Scale:**  $7(1, 12) = (7, 84)$ : ₹7 lakh of Nifty.
- **Add:**  $(1, 12) + (1, 7) = (2, 19)$ : ₹1 lakh in each.

### LINEAR COMBINATION

Scale vectors and add them:  $x c + y d$ . Here, a linear combination of the funds is simply a **portfolio**.

### STEP 3

## Matrix form: $Ax = b$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 10 \\ 105 \end{bmatrix}}_b$$

**Rows of  $A$**  = the rules (budget, return). **Columns of  $A$**  = the funds.

**By columns:**  $A \begin{bmatrix} 7 \\ 3 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 12 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}$

**By rows:**  $1 \cdot 7 + 1 \cdot 3 = 10, \quad 12 \cdot 7 + 7 \cdot 3 = 105$

#### KEY FACT

$Ax$  is a combination of the columns of  $A$ .

# Row picture: lines that cross

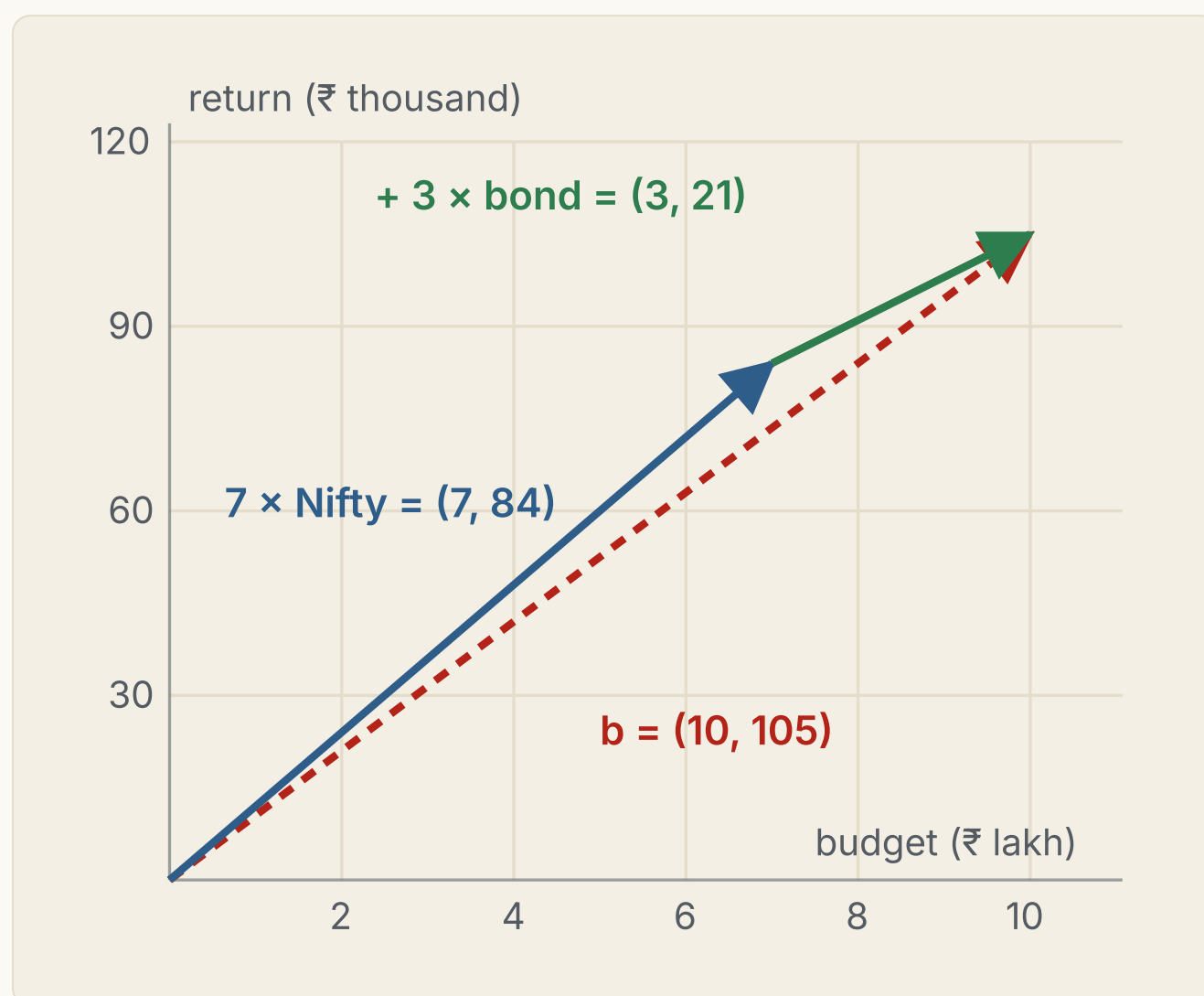


Each rule is a line. The answer is where they cross:  
**₹7 lakh Nifty, ₹3 lakh bonds.**

Check:  $7 + 3 = 10$  ✓ and  $84 + 21 = 105$  ✓

# Column picture: mix the arrows

$$x \begin{bmatrix} 1 \\ 12 \end{bmatrix} + y \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}$$



Same answer, seen as a **mix of fund arrows** landing exactly on the target  $b$ .

# Which targets can we reach?

The **span** of some vectors = every point you can reach by mixing them.

| Funds you may use | Span            | Meaning                |
|-------------------|-----------------|------------------------|
| Nifty only        | a line          | always exactly 12%     |
| Bonds only        | another line    | always exactly 7%      |
| Nifty + bonds     | the whole plane | any budget, any return |

COLUMN SPACE  $C(A)$

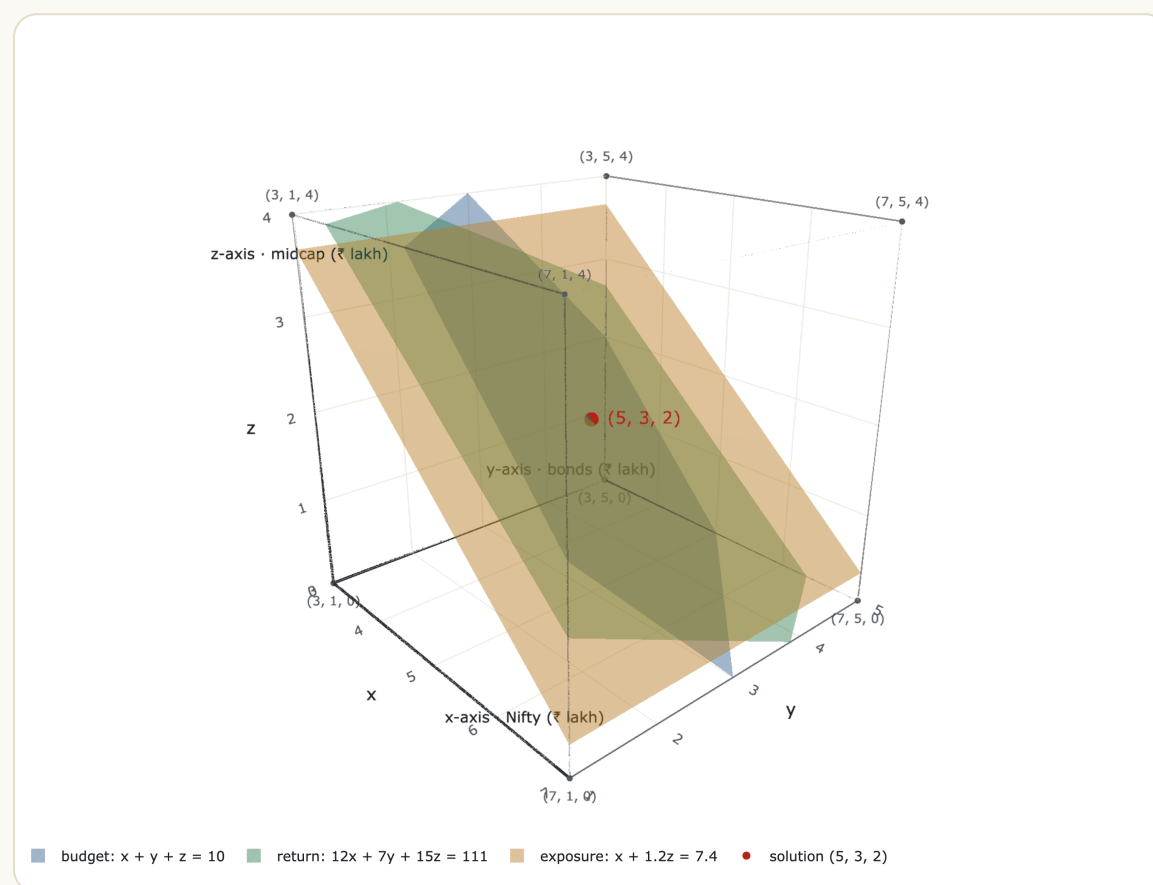
The span of the columns of  $A$ : every target your funds can produce.

WHY IT MATTERS

$Ax = b$  has a solution exactly when  $b$  is in  $C(A)$ .

# Add a midcap fund and a third rule

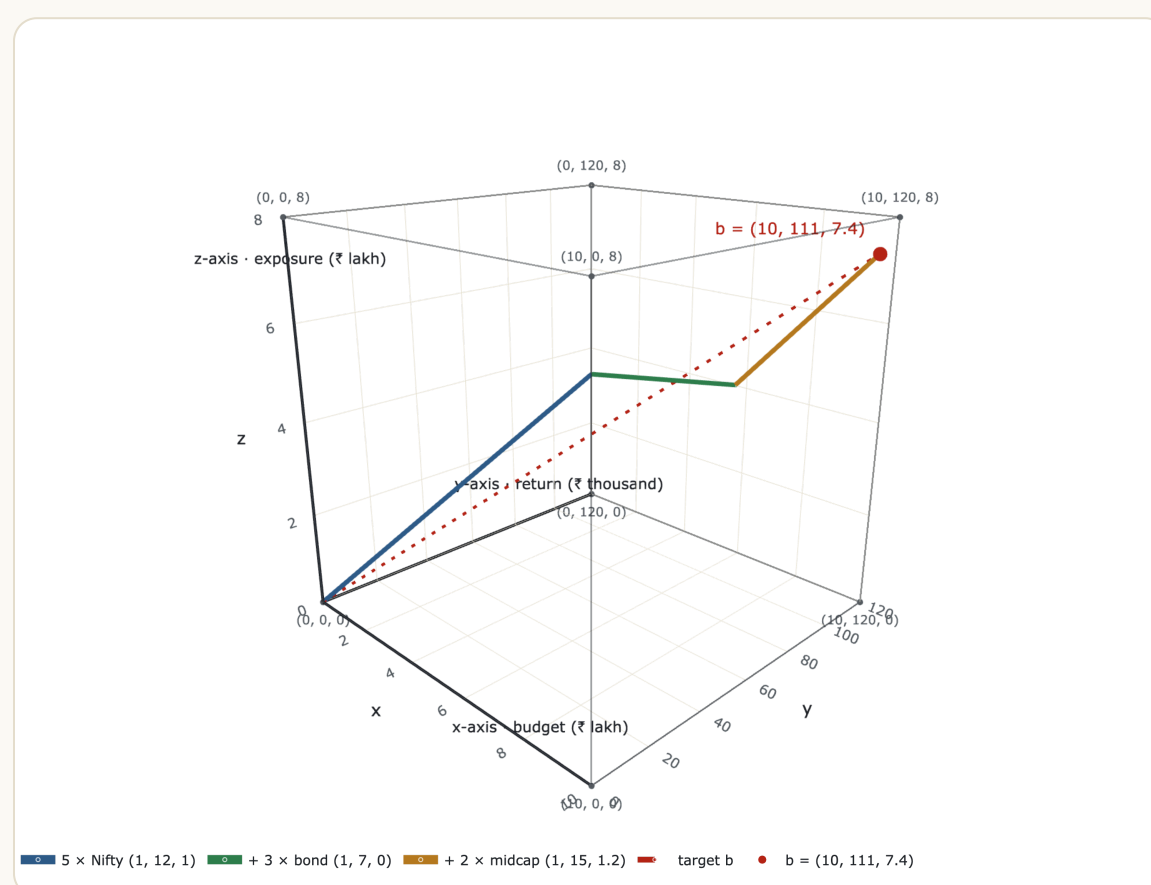
$$\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix} \quad \begin{array}{l} \text{budget} \\ \text{return} \\ \text{market exposure} \end{array}$$



Row picture: three planes meet at one point: ₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap.

# Column picture in 3D

$$5 \begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 15 \\ 1.2 \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix}$$



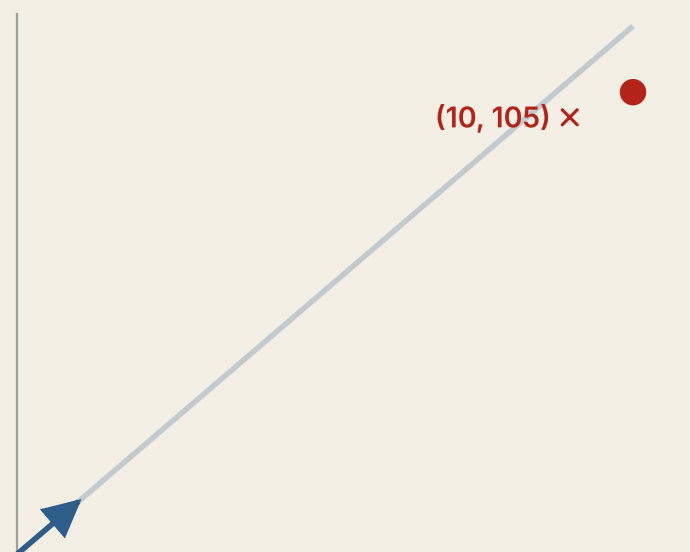
Change the target and the row picture gets three new planes. The column picture keeps the same three arrows, just mixed differently.

# Singular matrices

Two Nifty funds (both 12%) point the same way:



Parallel lines: no solution.



Every mix earns 12%: the target is off the line.

## SINGULAR VS INVERTIBLE

**Invertible:** the columns fill the space: one solution for every  $b$ . **Singular:** a fund adds no new direction (a twin fund, or a 50/50 balanced fund): no solution or infinitely many.

# Key takeaways

1. A linear system is  $Ax = b$ : rows are rules, columns are funds.
2. **Row picture**: lines (or planes) meet at the solution.
3. **Column picture ★**:  $b$  is a mix of the columns; the solution gives the amounts.
4.  $Ax$  is a combination of the columns of  $A$ .
5. Every target is reachable exactly when the columns fill the space ( $A$  invertible).
6. A fund built from others adds no direction:  $A$  is singular.

QUANT FROM FIRST PRINCIPLES

# The full interactive lesson is free

Fund mixer, 3D plots you can rotate, worked examples and practice:

**[learn.krisarj.com](https://learn.krisarj.com)**

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